

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. The body tissue that consists largely of material located outside of cells is
(A) epithelial tissue.
(B) connective tissue.
(C) muscle tissue.
(D) nervous tissue.
2. Which of the following would increase the rate of heat exchange between an animal and its environment?
(A) feathers or fur
(B) vasoconstriction
(C) wind blowing across the body surface
(D) countercurrent heat exchanger
3. Consider the energy budgets for a human, an elephant, a penguin, a mouse, and a snake. The _____ would have the highest total annual energy expenditure, and the _____ would have the highest energy expenditure per unit mass.
(A) elephant; mouse
(B) elephant; human
(C) mouse; snake
(D) penguin; mouse

Levels 3-4: Applying/Analyzing

4. Compared with a smaller cell, a larger cell of the same shape has
(A) less surface area.
(B) less surface area per unit of volume.
(C) the same surface-area-to-volume ratio.
(D) a smaller cytoplasm-to-nucleus ratio.
5. An animal's inputs of energy and materials would exceed its outputs
(A) if the animal is an endotherm, which must always take in more energy because of its high metabolic rate.
(B) if it is actively foraging for food.
(C) if it is growing and increasing its mass.
(D) never; due to homeostasis, these energy and material budgets always balance.
6. You are studying a large tropical reptile that has a high and relatively stable body temperature. How do you determine whether this animal is an endotherm or an ectotherm?
(A) You know from its high and stable body temperature that it must be an endotherm.
(B) You subject this reptile to various temperatures in the lab and find that its body temperature and metabolic rate change with the ambient temperature. You conclude that it is an ectotherm.
(C) You note that its environment has a high and stable temperature. Because its body temperature matches the environmental temperature, you conclude that it is an ectotherm.
(D) You measure the metabolic rate of the reptile, and because it is higher than that of a related species that lives in temperate forests, you conclude that this reptile is an endotherm and its relative is an ectotherm.

7. Which of the following animals uses the largest percentage of its energy budget for homeostatic regulation?
(A) marine jelly (an invertebrate)
(B) snake in a temperate forest
(C) desert insect
(D) desert bird
8. **DRAW IT** Draw a model of the control circuit(s) required for driving an automobile at a fairly constant speed over a hilly road. Indicate each feature that represents a sensor, stimulus, or response.

Levels 5-6: Evaluating/Creating

9. **EVOLUTION CONNECTION** In 1847, the German biologist Christian Bergmann noted that mammals and birds living at higher latitudes (farther from the equator) are on average larger and bulkier than related species found at lower latitudes. Suggest an evolutionary hypothesis to explain this observation.
10. **SCIENTIFIC INQUIRY** Eastern tent caterpillars (*Malacosoma americanum*) live in large groups in silk nests resembling tents, which they build in trees. They are among the first insects to be active in early spring, when daily temperature fluctuates from freezing to very hot. Over the course of a day, they display striking differences in behavior: Early in the morning, they rest in a tightly packed group on the tent's east-facing surface. In midafternoon, they are on its undersurface, each caterpillar hanging by a few of its legs. Propose a hypothesis to explain this behavior. How could you test it?
11. **SCIENCE, TECHNOLOGY, AND SOCIETY** Medical researchers are investigating artificial substitutes for various human tissues. Why might artificial blood or skin be useful? What characteristics would these substitutes need in order to function well in the body? Why do real tissues work better? Why not use the real tissues if they work better? What other artificial tissues might be useful? What problems do you anticipate in developing and applying them?
12. **WRITE ABOUT A THEME: ENERGY AND MATTER** In a short essay (about 100–150 words) focusing on energy transfer and transformation, discuss the advantages and disadvantages of hibernation.
13. **SYNTHESIZE YOUR KNOWLEDGE**



These macaques (*Macaca fuscata*) are partially immersed in a hot spring in a snowy region of Japan. What are some ways that form, function, and behavior contribute to homeostasis for these animals?

For selected answers, see Appendix A.